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Improving the accuracy of spectroscopic identification of geographical origins of agricultural samples through cooperative combination of near-infrared and laser-induced breakdown spectroscopy

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ABSTRACT

As a versatile strategy to improve accuracy for identification of the geographical origin of agricultural samples, milk vetch root samples in this study, both near-infrared spectroscopy (NIRS) and laser-induced breakdown spectroscopy (LIBS) have been cooperatively combined. The motivation was based on the potential of accuracy improvement by utilization of these two methods providing complementary spectral information, compositions of organic compounds and elements. For initial evaluation, NIRS and LIBS were separately employed to discriminate domestic milk vetch root samples from imported ones using support vector machine (SVM). The near-infrared (NIR) information in a full spectral range was used for the analysis, while in LIBS spectra, the intensities of 35 selected discrete element peaks were used. The use of NIR information providing organic compositions of the samples resulted in a discrimination accuracy of 91.5%, better than that of using LIBS elemental peak intensities (73.1%). Next, to utilize both sets of spectral data for discrimination, support vector regression (SVR) was used to represent NIR spectral feature of a sample as a SVR coefficient, and then it was merged with the existing discrete LIBS intensity data; accuracy was improved to 95.8%. The cooperative combination of information on organic and elemental composition of the samples was the root of improvement.

1. Introduction

International trading of agricultural products has been rapidly increasing; therefore, there is increasing chance of deception in regard to the origin of these products. Thus, an analytical method is needed to quickly screen the geographical origin of agricultural products. Vibrational spectroscopic methods such as near-infrared (NIR) and Raman spectroscopy are common choices [1–3] because they can quickly provide diverse chemical information on constituents of samples in a non-destructive manner. The compositions of agricultural products are complex, so their vibrational spectral features are severely overlapping with often indistinct signatures of individual components. Therefore, the use of chemometrics is generally necessary to extract relevant information from the spectra [1–3].

Meanwhile, another distinguishing feature of agricultural products could be the composition of elements, because this differs depending on cultivation conditions such as soil and water quality. Therefore, several previous studies determined the geographical origin of various samples such as saffron, citrus honey, wine, rice, coffee, date palm, and chili pepper [4–10] by utilizing their differences in elemental compositions using inductively coupled plasma-optical emission spectrometry (ICP-OES) or ICP-mass spectrometry (ICP-MS). On the other hand, laser-induced breakdown spectroscopy (LIBS) [11,12] is another type of atomic emission spectroscopy enabling more rapid and simpler elemental analysis. In LIBS measurements, a short laser pulse simply irradiates the sample surface to directly generate a micro-plasma for atomization and excitation of elements; therefore, it offers fast analysis with no or minimal sample preparation and broad elemental coverage including light elements. Therefore, LIBS-based identification or classification of geographical origins of soil, wine, hand-made cigars, and edible salts have been also demonstrated [13–16]. Moreover, a recently developed hand-held LIBS makes the analysis readily expandable for on-site measurements [17].

Since both vibrational spectroscopy and LIBS, individually characteristic methods for discrimination of various samples, have been separately evaluated for applications such as the discrimination of good

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coffee beans from those with defects using NIR and LIBS [18], the differentiation of aromatic plants using Raman, NIR and LIBS [19], and the determination of magnesium stearate using NIR and LIBS [20]. In addition, the combination of LIBS and vibrational spectroscopy that would enhance accuracy of a target analysis has been recently attempted, for example, the connection of LIBS and Raman spectroscopy for multivariate classification of pigments/inks [21] and bacteria [22]. As described, this attempt is in the beginning stages and so worth further exploration.

In this study, combination of vibrational spectroscopy and elemental analysis was evaluated to discriminate the geographical origin of milk vetch root (*Astragalus membranaceus*) samples. It is a major medicinal herb commonly used in formulations of oriental herbal medicines due to its cardio-protective and anti-inflammatory activities [23]. Therefore, it has been largely traded in Asian international market and the identification of its geographical origin becomes more important to prevent false designation of origin. Initially, NIRS and LIBS were independently used for discrimination using support vector machine (SVM) [24]. Because NIR bands of the samples were broadly present in a whole spectral range due to the combination and overtone vibrations, entire spectral information could be used. In the case of LIBS data, the intensities of 35 selected element peaks were used rather than the whole spectrum because element peaks were discrete with no serious overlap. Discrimination errors obtained in the two cases were then compared to each other.

Next, one important issue to consider was how to combine NIR and LIBS data of significantly different spectral characteristics, such as broad NIR bands and narrow non-overlapping LIBS peaks, for discrimination. Because LIBS peaks are discrete, it would be better to represent overall NIR spectra as a discrete feature-carrying variable and combine it with the LIBS variables. For this purpose, support vector regression (SVR) [25] was used to represent each complex NIR spectrum as a discrete SVR coefficient. Then, the coefficient was incorporated with the LIBS variables for discrimination, and subsequent accuracy was evaluated. Finally, sensitivity analysis was performed to examine the contributions of NIR and LIBS variables on the achieved discrimination accuracy.

2. Experimental section

2.1. Preparation of milk vetch root samples

All milk vetch root samples (25 domestic and 25 imported each) were kindly provided in powder form by the National Agricultural Products Quality Management Service (NAQS, an official test organization) in Gimcheon, Korea. The samples cultivated in diverse regions with various climates were cautiously collected by the NAQS to include wide compositional variation in the dataset and make the dataset as representative as possible. The samples were acquired during harvest season. At each cultivation area, no duplicate sampling was accomplished. Pellets were prepared by placing 0.3 g of the powder sample into a 13-mm-diameter stainless-steel die under a pressure of 12 ton/cm^2 (PIKE, CrushIR Digital Hydraulic Press). The thickness of the pellets was approximately 2.0 mm.

2.2. Acquisition of LIBS and NIR spectra

LIBS spectra of the samples were collected using a commercial instrument (J200 Tandem LA-LIBS Instrument, Applied Spectra, Inc.) equipped with a Nd:YAG laser (maximum energy: 25 mJ) generating 266 nm (a harmonic wavelength) with a pulse width of 5 ns. The instrument was able to collect spectra with a resolution of 0.10 nm in a 185–1048 nm range using a six-channel CCD. The laser power was maintained at 17.5 mJ for sample measurement, and the diameter of laser illumination was approximately 100 μm.

Initially, by changing the delay time in spectral acquisition after laser excitation from 0.0 to 1.4 μs, LIBS spectra (185–1048 nm range) of one domestic sample were collected. (refer to Fig. S1 in the supplementary data). When the delay time was zero or short (e.g., 0.2 μs), the background occurring from continuous emission from recombination and Bremsstrahlung radiation [11] was dominant as expected. The background obviously obscures the analysis of element peaks. Meanwhile, the magnitude of the background substantially decreased as the delay time increased to 0.4 μs and no substantial change was observed after a delay time of 0.5 μs. So, the delay time of 0.5 μs was chosen for subsequent experiments to clearly ensure no presence of the background in LIBS spectra.

With the delay time of 0.5 μs, LIBS spectra of four different samples (two imported and two domestic samples) were collected by randomly irradiating the laser to 25 different spots on a sample. Triplicate spectra were collected at each spot. Then, relative standard deviations (RSD) of intensities of Ca (393.34 nm), K (766.43 nm), Mg (518.36 nm), and Na (588.97, 589.59 nm) peaks were evaluated as the number of measured spots included in the spectral averaging increased (refer to Fig. S2 in the supplementary data). For example, in the averaging of three-spot measurements, 2300 combinations ($_{25}C_3$, chances for selection of 3 spots out of all 25 spots) were available, while there were 53,130 ($_{25}C_{20}$) cases in the averaging of 20 different spots. Out of all the possible combinations, only 100 combinations in each case were randomly selected to evaluate the reproducibility. As expected, in each case, RSDs decreased as the larger numbers of spots were sampled to obtain LIBS spectra and were improved further when more than 25 spots were averaged. To maintain balance between reproducibility and spectral acquisition time, the use of six spot-averaged LIBS spectra yielding around 5.0% RSDs (5.4% for K, 5.8% for Mg, 5.5% for Na) was decided for subsequent analysis; while, the RSD of Ca peak was slightly higher (7.0%).

Diffuse reflectance NIR spectra (9000–4000 cm^{-1}) of the same pellet samples were collected using an FT-NIR spectrometer (NIRFlex N-500, BUCHI). A pellet made from Teflon powder was used as a background. An inteferogram was converted to a singlebeam spectrum using a Boxcar function without zero filling. NIR spectral acquisition was performed before the collection of LIBS spectra. For each sample, six replicate spectra (three spectra from one face and three spectra from the other face) were collected with a resolution of 4 cm^{-1}.

All raw NIR spectra were treated with multiplicative scatter correction (MSC) [26] to reduce unwanted spectral variations by scattering. The MSC model introduced by Marten et al. [27] which compensating both offset and amplification was used. The detail equation and relevant description are provided in the supplementary data. Also, area normalization was performed for all LIBS spectra before discriminant analysis. For the normalization, the area in the 185–1048 nm range was initially calculated and then individual intensities in a spectrum were divided by the area. MSC, area normalization, and principal component analysis (PCA) were conducted using Matlab, version 7.0 (The MathWorks Inc., Natick, MA, USA).

2.3. SVM, SVR, and Sobol's variance-based sensitivity analysis

SVM, SVR, and Sobol's variance-based sensitivity analysis were accomplished using R, version 3.2.0 [28,29]. SVR is a regression model derived from SVM, a binary classifier, that estimates the curved-surface closest to data. SVR is set to increase the value of the ε – insensitive loss function as each data point moves away from the regression surface, and, to optimize it, the Lagrangian algorithm is used to make the curvature as close to all data as possible.

SVM is a widely-used data mining technique to solve the binary classification problem. SVM maximizes the margin, which is the distance to two classes, and the estimated plane maximizing the margin is called the hyperplane. Since many experimental data such as NIR and LIBS spectra are difficult to classify into simple planes, SVM classifies data using kernel functions, which are nonlinear methods. Thus y is

used for classification with input data $\boldsymbol{x}$, SVM can be written as follows.

$$y = K(w, \boldsymbol{x}) + b$$

where K is the kernel function and b is the bias. The norm of w is the margin from the hyperplane to the data.

After SVM, variance-based sensitivity analysis (as known by Sobol's method [30]) was performed to measure the influence of the input variables on the classification results of the model. Sobol's method is one of the typical Monte-Carlo simulation-based sensitivity analyses, which calculates the total effect indices S_{T_i} of each input variables. The total effect index S_{T_i} is a measure of the relationships between input variables and uncertainty of output variable (expressed as variance) which defined as follows.

$$S_{T_i} = \frac{E(V(Y \mid X_{\sim i}))}{V(Y)} = 1 - \frac{V(E(Y \mid X_{\sim i}))}{V(Y)}$$

where E(·) means expectation, V(·) means variance, X_i is the ith input variable, $X_{\sim i}$ is the all input variables except X_i, and Y is the output variable expressed as a result of classification. A larger value of S_{T_i} means that the ith input variable contributes significantly to the performance of the classification model. For a more detailed description of Sobol's method, refer to the supplementary data.

3. Results and discussion

3.1. Discrimination of geographical origins of milk vetch root samples by NIRS

Fig. 1 shows the average diffuse reflectance NIR spectra of domestic (blue) and imported (red) milk vetch root samples in the 9000–4000 cm^{-1} range. The same-color shade superimposed with each average spectrum corresponds to the standard deviation, that indicating the magnitude of spectral variation among the samples. The 8500–7700 and 7200–6000 cm^{-1} ranges, where the spectral differences between two groups were relatively large, are highlighted in the insets. The overall spectral shapes of each group were similar because major chemical compositions of the samples from both geographical origins are not considerably different. On the other hand, small compositional and structural dissimilarities among the components brought the spectral differences (see the insets of Fig. 1), which were valuable in discriminating between samples.

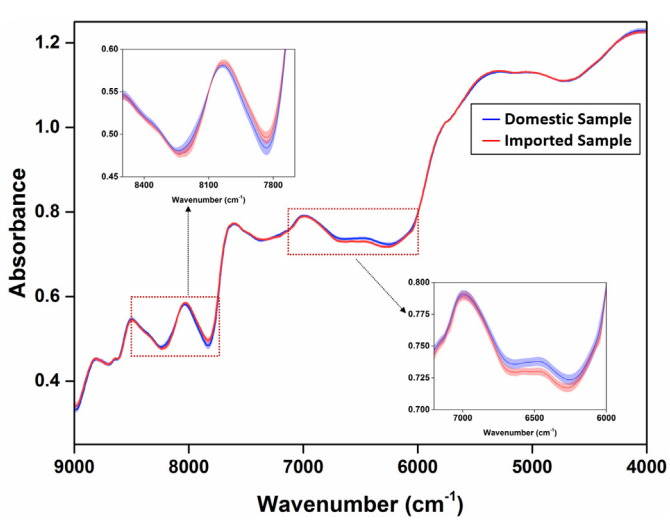

Fig. 1. Average diffuse reflectance NIR spectra of domestic (blue) and imported (red) milk vetch root samples in the 9000–4000 cm^{-1} range. The same-color shade superimposed with each average spectrum corresponds to the standard deviation, which reflects the magnitude of spectral variation among the samples. The 8500–7700 and 7200–6000 cm^{-1} ranges are magnified in the insets. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

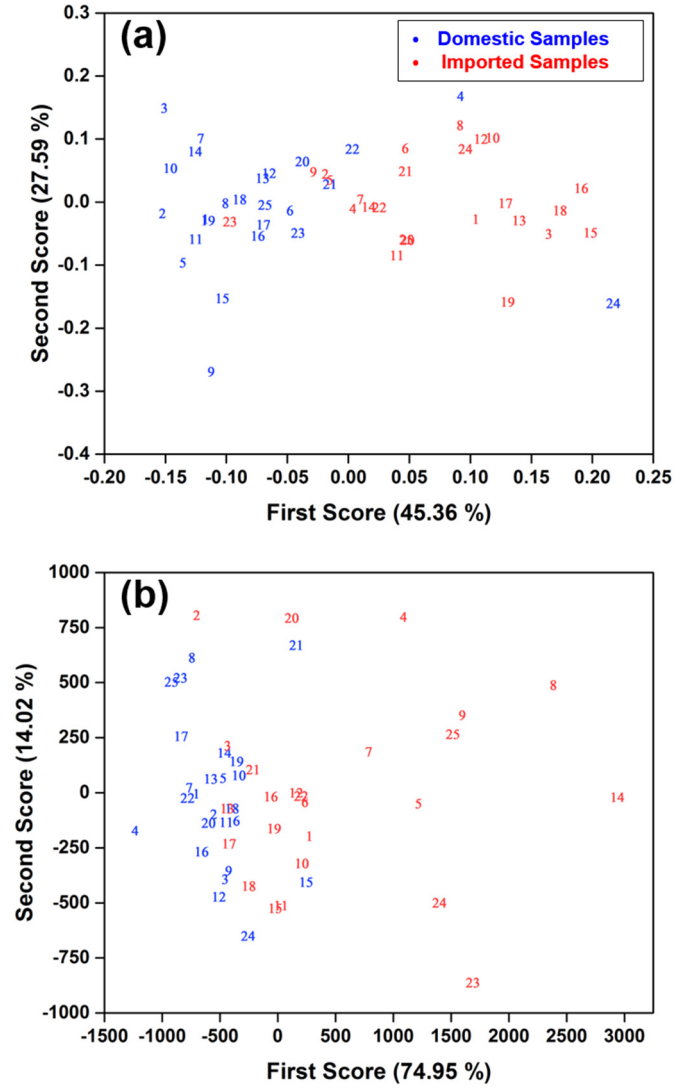

Fig. 2. PCA score scatter plots constructed using the first and second scores obtained from the NIR (a) and LIBS (b) spectra of the samples.

To visualize the overall differences among all of the samples, PCA was performed using all MSC-treated NIR spectra of the samples, and the obtained scores were examined. Fig. 2 (a) shows the corresponding score scatter plot constructed using the first and second scores. The corresponding loadings are also shown in the supplementary data (Fig. S3). The numbers correspond to the sample numbers (25 samples in each group). Two groups of domestic (blue) and imported (red) samples are identifiable, though they are partially overlapping with each other. Therefore, some degree of error exists when discriminating the samples. One notable observation is that the #24 domestic sample is located at the far-right in the imported sample group. It was interesting to see where this sample was located in the score domain constructed using LIBS spectra.

Using all variables in the full spectral range, accuracy for the discrimination of domestic and imported samples using SVM was evaluated. For SVM-based analysis, it is necessary to determine an optimal kernel function, as well as optimal parameters of a kernel and the cost parameter for sensitivity of a SVM. For this purpose, three kernels were tested. The radial kernel with sigma from 1 to 5, polynomial kernel with degree from 1 to 5 and scale from 0 to 4, and linear kernel functions were tested simultaneously. Also the same cost parameters of 1, 10,

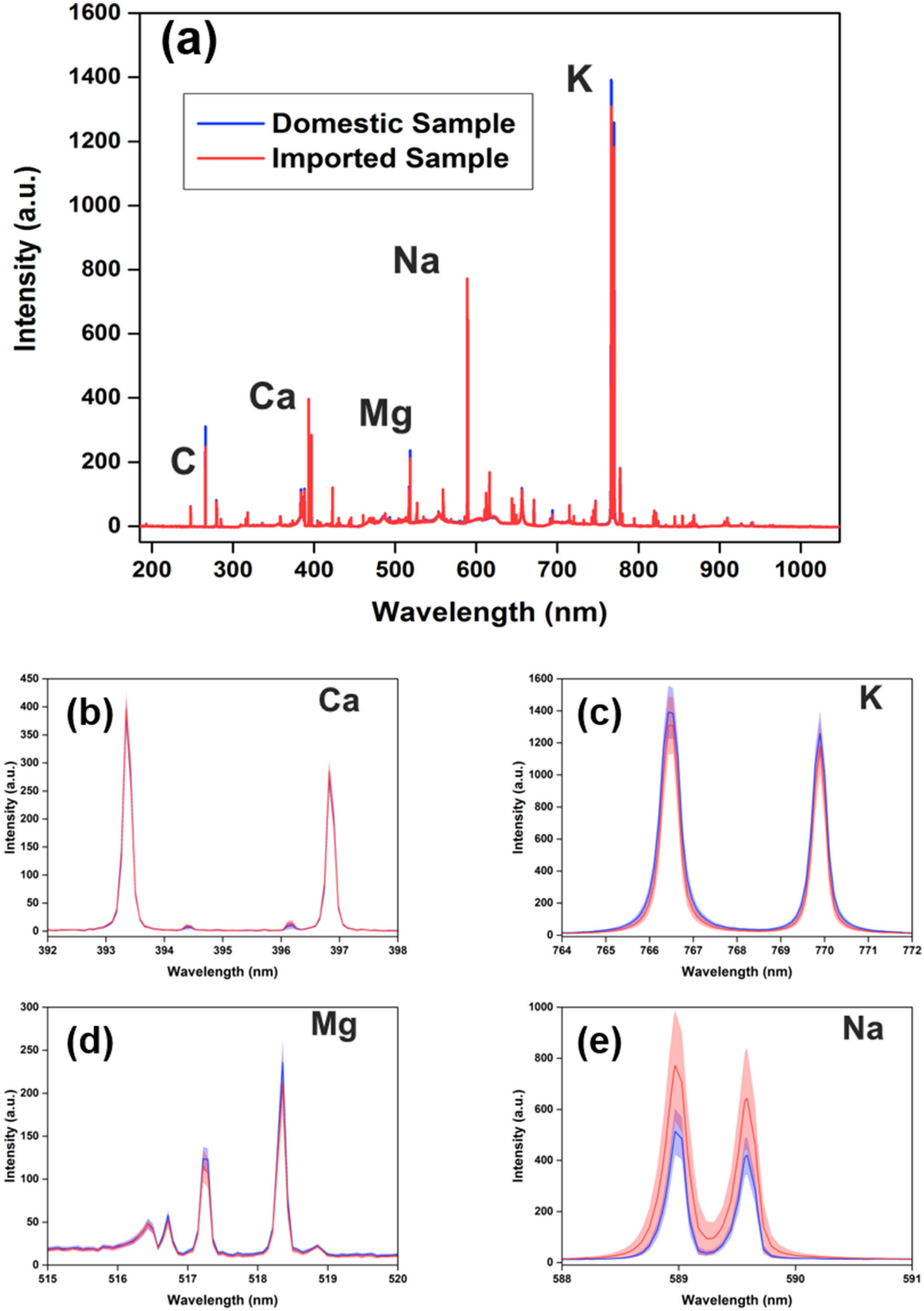

Fig. 3. Average LIBS spectra of all domestic (blue) and imported (red) milk vetch root samples in the 185–1048 nm range (a). For detailed examination, the peaks of the four main elements are magnified ((b) through (e)). Note that two small peaks at 394.37 and 396.19 nm (b) correspond to Al. The shades around the spectra correspond to the standard deviation, which indicates the variation in the element concentration among the samples. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1
Wavelengths of 35 selected element peaks for SVM and the corresponding elemental information.

Wavelength (nm)	Element	Intensity	Wavelength (nm)	Element	Intensity
247.83	C I	Middle	526.99	Ca I	Middle
279.51	Mg II	Middle	558.88	Ca I	Middle
280.27	Mg II	Middle	559.44	Ca I	Middle
383.25	Mg I	Middle	559.81	Ca I	Middle
383.82	Mg I	Middle	588.97	Na I	High
385.03	CN (0,0)	Middle	589.59	Na I	High
385.43	CN (1,1)	Middle	610.30	Ca I	Middle
386.16	CN (2,2)	Middle	612.20	Ca I	Middle
387.12	CN (3,3)	Middle	616.21	Ca I	Middle
388.32	CN (4,4)	Middle	643.95	Ca I	Middle
393.34	Ca II	High	646.28	Ca I	Middle
394.37	Al I	Low	656.33	H α	Middle
396.19	Al I	Low	714.85	Ca I	Middle
396.82	Ca II	High	746.88	N I	Middle
422.69	Ca I	Middle	766.43	K I	High
516.72	Mg I	Middle	769.90	K I	High
517.22	Mg I	Middle	777.20	O I	Middle
518.36	Mg I	Middle			

CN: CN violet system (CN molecular emission).
H_α: Deep-red visible spectral line in the Balmer series.
I and II indicates oxidation numbers of the corresponding elements.
The low, middle, and high intensity corresponds to below 50, 50–300 range, and greater than 300, respectively.

100, and 1000 were tested for three kernels. To find optimal SVM parameters, cross-validation of 1000 times was performed by randomly assigning 20 samples from each group for training and the remaining 5 samples for validation. The SVM using the linear kernel with the cost parameter of 1000 provided the best discrimination accuracy, 91.5% with a standard deviation (SD) of 7.3%. The spectral differences shown in Fig. 1 were the sources enabling differentiation between the domestic and imported samples with good accuracy.

Although a metabolomics approach to characterize two different species of milk vetch roots using AFLP based genetic fingerprinting and GC–TOF/MS-based metabolic fingerprinting was reported [31], provision of a justification explaining the observed spectral differences for all the employed samples should require extensive phytochemical investigation with full compositional analysis using destructive analytical methods and was thus beyond the scope of this study. Nonetheless, the differences would be originated from genes, metabolic pathways [31], cultivation conditions, and/or other factors.

3.2. Discrimination of geographical origins of milk vetch root samples using LIBS

Under the optimized conditions determined in the experimental section, LIBS spectra of all 25 domestic and 25 imported milk vetch root samples were acquired. Fig. 3 (a) shows average LIBS spectra of all domestic (blue) and imported (red) milk vetch root samples in the 185–1048 nm range. As seen, the peaks of Ca, K, Mg, and Na were dominant, and overall spectral features between the two average spectra were similar. Although diverse element peaks appeared in the spectra, there were no specific element peaks that were only presented either in the domestic sample spectra, or only present in the imported sample spectra. This observation informs us that the elemental compositions of milk vetch root samples from both origins were similar.

For detailed examination, the peaks of the main four elements are separately magnified in Fig. 3 (b) through (e). The shades around the spectra correspond to the standard deviations, indicating the variations of element concentration among the samples. The concentration of Na was different between groups, and this dissimilarity can be utilized for discrimination. However, the magnitude of the standard deviation was substantially greater for the imported samples, so the standard deviations of the groups were largely overlapping. Therefore, the distinct identification of the two groups by the content of Na would not be possible. In the cases of K and Mg, the average peak intensities of the two groups were somewhat different, but the corresponding standard deviations were overlapping. The dissimilar contents of K and Mg in the samples could be additionally helpful for discrimination, although the differences in Na content were the most distinguishing. Furthermore, the Ca content in the samples of both groups was similar.

Fig. 2 (b) shows the scores obtained from all LIBS spectra. The corresponding loadings are also shown in the supplementary data (Fig. S3). The scores were more overlapped with each other compared to those obtained from the NIR spectra and the distribution of the imported samples was broader, probably due to the larger variation in Na content (Fig. 3 (e)). The domestic sample #24 (blue color), located inside the opposite sample group in the NIR score domain, moved into the domestic sample group. A similar observation was observed for the imported sample #23 (red color). This observation suggests that the information provided by each spectroscopic method was independently characteristics, and the combination of both information could improve discrimination accuracy. In a meanwhile, the improvement would be dependent on organic and elemental compositions of samples.

Because the element peaks in the LIBS spectra are discrete, use of information in the whole spectral range including a large portion of baseline containing noise or very minute elemental information is not desirable for discriminant analysis. Therefore, 33 peaks with intensities greater than 50 were selected. In addition, two Al peaks were selected although their intensities were below 50. Table 1 summarizes the positions of 35 selected peaks and the corresponding elemental information. SVM-based discrimination accuracy using the 35 selected variables was 73.1% ± 10.8%. The linear function with the same parameters was used in SVM. The differences in the concentrations of elements such as Na were attributed to the obtained discrimination accuracy; while, the accuracy was inferior to that of NIR-based analysis. It demonstrates that the information on organic constituents of the samples provided by NIR spectroscopy was more discriminant. When the whole spectrum without the selection of the 35 element peaks was used, the accuracy decreased to 68.7%. It indicates that the intensities of many minor peaks and noisy baseline (excepting the intensities of 35 peaks) make the main element peaks blurred in SVM-based discrimination. Therefore, the differences in the concentrations of the selected elements would be cooperatively useful for further improvement of accuracy when they are combined with the NIR information.

Peak intensity of an element does vary according to its concentration. Nonetheless, to assess the correlation, three samples yielding high Na peak intensities and three samples with low Na peak intensities were selected, and the Na concentrations in these samples were determined using ICP-OES. The correlation between the determined Na concentrations of the samples and corresponding Na peak intensities is shown in Fig. S4 in the supplementary data. The resulting R^2 was 0.99, demonstrating that Na peak intensity is associated with concentration.

3.3. Combination of NIR and LIBS spectral information for discrimination

For the combination of both spectroscopic datasets, it could be appropriate to represent the whole NIR spectrum as a single variable to combine with discrete LIBS variables. For this purpose, SVR was initially used to represent each NIR spectral spectrum as a discrete feature-carrying variable (SVR coefficient). SVR is a data mining technique that represents surface data. If $\boldsymbol{x}$ is defined as NIR spectrum and y is defined as class of import or domestic samples, then SVR can be written as (1).

$$y = w \bullet \boldsymbol{x}_{NIR} + b \text{ with } w \in X,\ b \in R \tag{1}$$

$$y = \begin{cases} 1 & \text{if domestic sample} \\ -1 & \text{if imported sample} \end{cases}$$

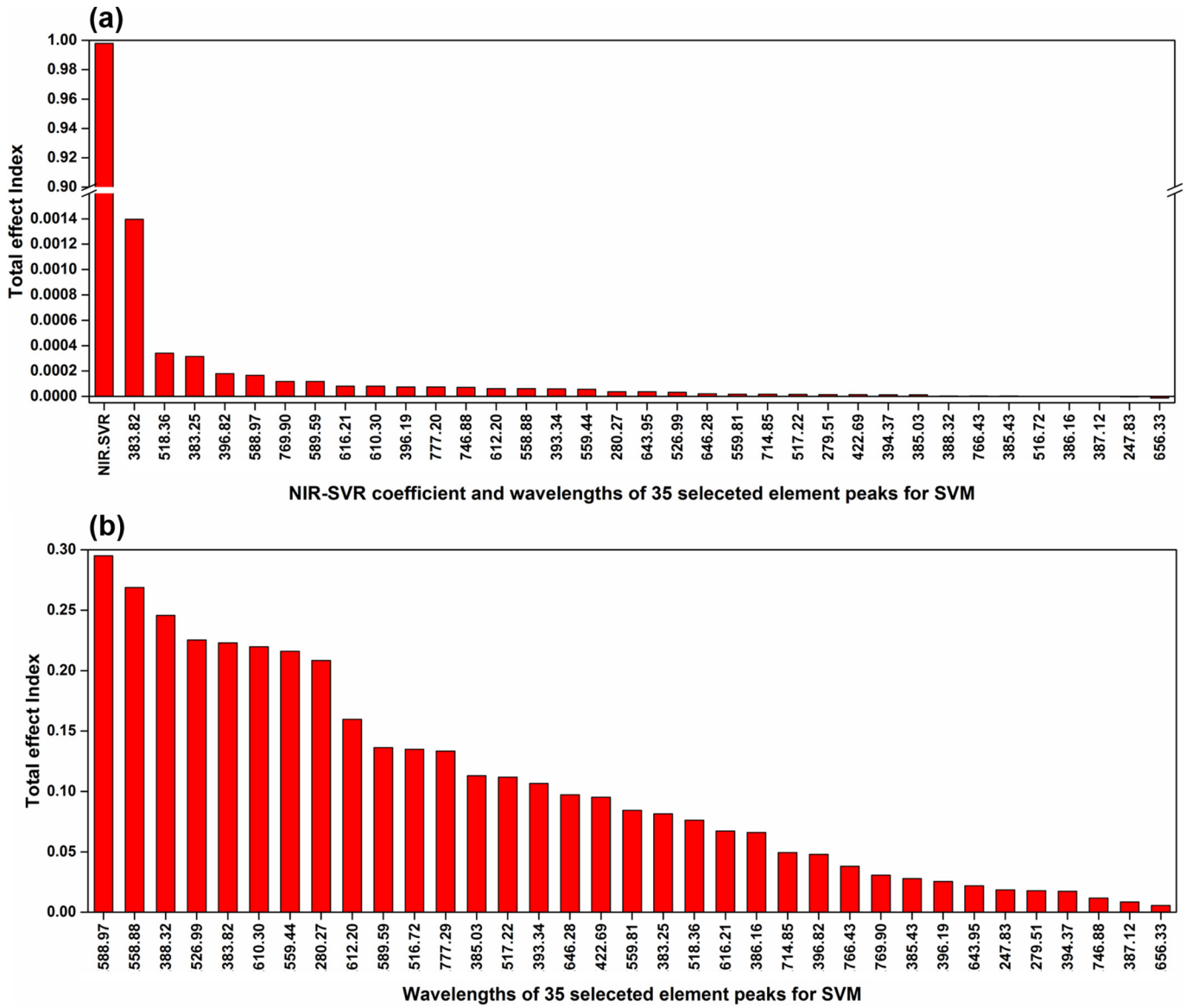

Fig. 4. Result of Sobol's variance-based sensitivity analysis showing the calculated total effect indexes for 36 selected variables including one NIR-SVR coefficient and 35 LIBS peak intensities (a). The sensitivity analysis performed using only the 35 LIBS variables is also shown (b).

where w is the weight for the NIR spectrum, b is the bias of the model and '·' is the inner product. The norm of the w is the distance from the plane to the data and the SVR estimates the weight that minimizes the distance in (2).

$$minimize \quad \frac{\|\boldsymbol{w}\|^2}{2} \tag{2}$$

Most of experimental data are difficult to explain in a plane, and data are more often better represented by curved surfaces, which are nonlinear models. Optimization is performed using the kernel function to obtain the surface in SVR. We tried to linear, radial and polynomial kernels, and the linear kernel was used for analysis of NIR by comparing the mean square error (MSE). Thus (1) can be rewritten as (3).

$$y = K(w, \boldsymbol{x}_{NIR}) + b \tag{3}$$

where $K(w,\boldsymbol{x})$ is the kernel function. It also applies penalties to data away from the SVR surface using a slack variable ξ_i, ξ_i^* and flexibly changes the penalty using the constant C. Various values were applied to determine the constant C and a constant of 1000 was provided.

$$minimize \quad \frac{\|w\|^2}{2} + C\sum_{i=1}^{l}(\xi_i + \xi_i^*) \tag{4}$$

To optimize (4), the Lagrangian and dual problem solver are used, then $\widehat{w}$ and $\widehat{b}$ are obtained. For more detailed descriptions on SVR as well as SVM are provided in the supplementary. Using the NIR spectrum, the fitted value (NIR-SVR coefficient) $\widehat{y}$ can be calculated from (5).

$$\widehat{y} = K(\widehat{w}, \boldsymbol{x}_{NIR}) + \widehat{b} \tag{5}$$

The obtained NIR-SVR coefficient was then merged with the existing LIBS variables as the 36th variable for discrimination. With using these variables, the resulting discrimination accuracy was improved to 95.8% ± 4.1%. Also, the linear function with the same parameters was used in SVM, and the improvement in accuracy was statistically significant based on F-test (95% significant level, p-value < < zero).

To investigate how significantly each variable contributes to the discrimination, Sobol's variance-based sensitivity analysis [30] was performed. This method provides a total effect sensitivity index (or total effect index), which indicates the influence of each variable on the

variation of the target analysis result as a quantitative number between 0 and 1. A variable yielding a larger total effect index implies a greater contribution to the target analysis. A more detailed description of Sobol's variance-based sensitivity analysis is included in the supplementary data.

Fig. 4 (a) presents the results of the sensitivity analysis showing the calculated total effect indexes for 36 selected variables. The first variable is the NIR-SVR coefficient, and the other variables are derived from the selected 35 LIBS peaks at diverse wavelengths. The total effect index of NIR-SVR coefficient was 99.8%. This indicates that NIR information plays a predominant role in discriminating the domestic and imported milk vetch root samples. The variable yielding the second greatest total effect index was the peak at 383.82 nm, corresponding to Mg, while its contribution (0.14%) was much smaller than that of the NIR-SVR coefficient. The peaks yielding the third and fourth greatest total effect indexes were at 513.36 nm and 383.25 nm, also originating from the emission of Mg. The Ca peak at 396.82 and the Na peak at 588.97 nm ranked fifth and sixth, respectively, in the order of total effect index. Although it was expected that Na information would largely contribute to the discrimination as shown in the LIBS spectra (Fig. 3), Na information might not be second in influence when both data were combined. In a meanwhile, the Na intensity at 588.97 nm contributed most to the discrimination (approximately 30%) when only 35 LIBS variables were used (refer to Fig. 4 (b)). In the absence of NIR information, the largely varying Na intensity was highly valuable for discrimination.

4. Conclusion

Compared to the use of either NIRS or LIBS alone, the incorporation of both spectral features was effective in improving accuracy for the discrimination of geographical origins of milk vetch root samples. The representation of whole NIR spectral feature into a single SVR coefficient and merge it with discrete LIBS peak intensities for discriminant analysis was unique strategy proposed for the combination of NIRS and LIBS information. The combination of other vibrational spectroscopic methods such as Raman spectroscopy able to provide more-component specific spectral features with LIBS is also promising; therefore, it is worthwhile to explore further diverse applications such as vibrational spectroscopy-based biomedical diagnosis and jewelry appraisal, in which the compositions of both organic constituents and elements in the samples would provide supplementary information for a target analysis.

Acknowledgment

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.sab.2018.09.004.

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